

Network Imaging Package for Sapera LT

Optimization Guide

Edition 5.00

sensors | cameras | frame grabbers | processors | **software** | vision solutions



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Teledyne DALSA offers the widest range of machine vision components in the world. From industry-leading image sensors through powerful and sophisticated cameras, frame grabbers, vision processors and software to easy-to-use vision appliances and custom vision modules.

Teledyne DALSA is based in Waterloo, ON, Canada, the company has operations in Montreal, QC; Bromont, QC; Eindhoven, NL; Munich, Germany and Tokyo, Japan.

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Introduction

This document describes the Sapera Network Imaging Package and how to modify the network adapter (NIC) settings to optimize performance for different application requirements, such as:

- Increase Throughput
- Quick Response and Low Latency
- Throughput
- System Robustness
- Lower CPU Utilization

Information on optimizing certain system settings, such as the Processor Power Management parameters available with 6th generation Intel CPU chips is also provided.

In general, the goal is make the system as close to a real-time operating system as possible by reducing the number of applications and services running in the background, as well as eliminating any unnecessary network traffic unrelated to GigE Vision operation and the imaging application.

The document organizes material into the following sections:

- Sapera Network Imaging Package Overview
- Managing Network Connections
- Network Adapter Optimization
- Camera Features Controlling Image
- Host Computer System Considerations

For more information, such as troubleshooting, refer to the Sapera LT Getting Started Manual for GigE Vision Cameras & 3D Sensors.

Sapera Network Imaging Package Overview

Description

The **Teledyne DALSA Network Imaging Package** contains the underlying networking components necessary to control and acquire images from GigE Vision compliant cameras. It is included with Sapera LT 8.01 or later and installed on the host computer during installation. The user selects to install the package via a tick box show during the Sapera installation.

The package includes:

- **Network Configuration Tool:** User control for GigE Vision networking parameters. Tool provides information on all network adapters installed in the system and any connected GigE Vision cameras, as well as camera recovery.
- **Status Window Tool:** Tray utility providing status and information on any connected GigE Vision device.
- **Smart DHCP Server:** available if no DHCP server is available on the network.
- **GigE Server:** Underlying server interfacing GigE Vision devices with the host computer, using the GigE Vision Control Protocol (GVCP).
- **GigE Vision Driver:** Provides the interface to Sapera LT for control and image acquisition.
- **Sapera Filter Driver:** Underlying driver managing data transfers from the GigE Vision device. Filters network traffic for image packets only.

Default settings typically are adequate for GigE Vision camera installations. This manual describes the Teledyne DALSA Network Configuration Tool and how to fine tune network settings to optimize camera installations for differing applications.

Supported Industry Standards



100% compliant with the GigE Vision 1.0 specification which defines the communication interface protocol used by any GigE Vision device. The device description and capabilities are contained in an XML file. For more information see:
<http://www.machinevisiononline.org/public/articles/index.cfm?cat=167>

The GigE Vision specification of the Automated Imaging Association (AIA, www.machinevisiononline.org) defines the communication interface protocol used by any GigE Vision device.

GenICam specification of the European Machine Vision Association (EMVA, www.emva.org). Defines the capabilities of any GigE Vision device.

Application Development in a GigE Vision Compliant Environment

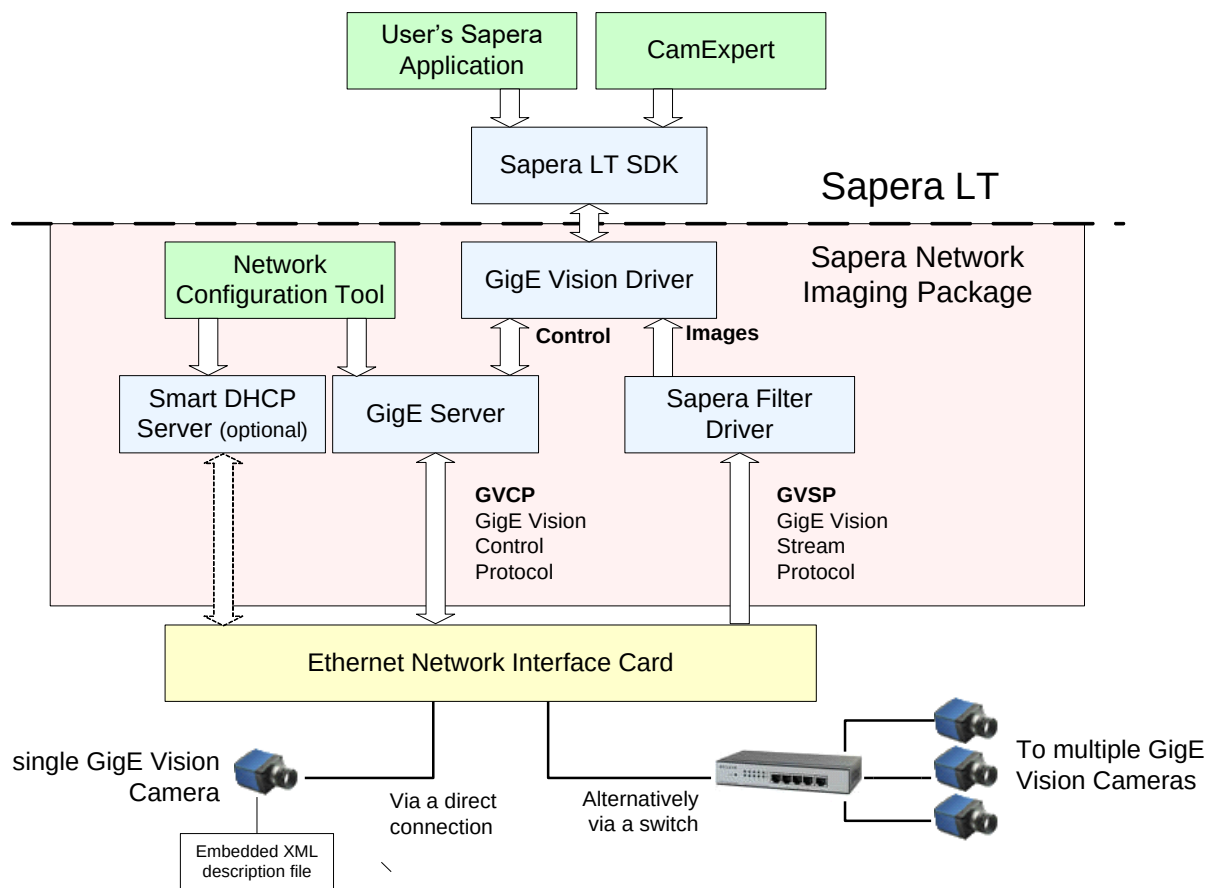
The following graphic is the functional block diagram of a Spera application with GigE Vision cameras. The Spera Network Imaging Package installation includes the GigE Vision driver and the Spera Filter Driver.

When using a Teledyne DALSA camera, the GigE Vision Compliant XML description file is embedded within the camera firmware allowing GigE Vision Compliant applications to know camera capabilities immediately after connection.

The GigE Server provides the control channel to the camera for sending commands to change features using the GigE Vision Control Protocol (GVCP).

The camera uses the GigE Vision Stream Protocol (GVSP) to construct image data packets to send to the NIC.

The Spera Filter Driver checks all incoming packets from the NIC to determine if they contain image data. If so, the packet is forwarded to the GigE Vision driver, otherwise it is passed through to the host.

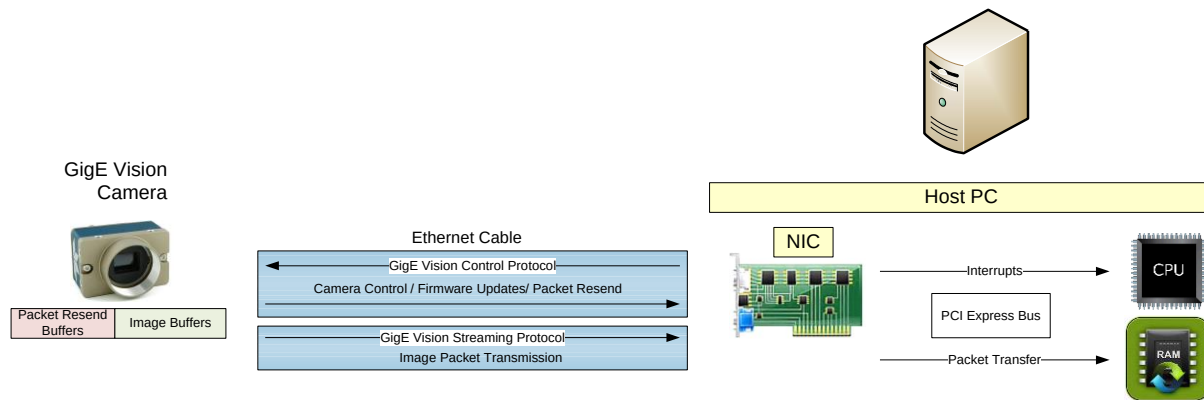


Overview

Implementation

When an Ethernet packet sent from the camera is received by the NIC, the packet is stored in the receive data FIFO on-board memory. Depending on the particular NIC, on-board memory size can differ (for example, 64KB, 256KB or higher). It is then transferred to a receive buffer in host memory.

The NIC then sends an interrupt request (IRQ) to alert the CPU (or CPU core) that a packet has arrived in host memory. At this point, packets containing image data are filtered by the Sapera Filter Driver. When the Sapera image buffer is complete, the Sapera application is notified that the buffer is available.

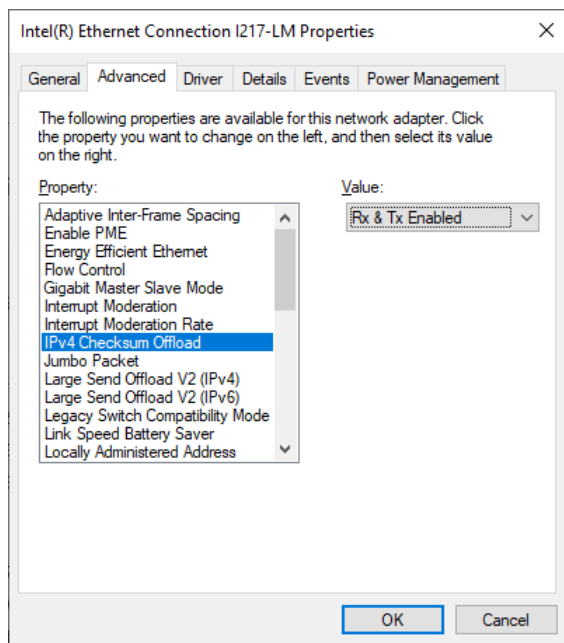


Checksums

Checksums are used to ensure the integrity of data portions for data transmission or storage. The transmitter will calculate a checksum of the data and transmits the data together with the checksum. With the UDP protocol, when a packet is received, the checksum is verified; if a checksum fails, the packet is dropped. When the Sapera GigE Driver discovers that a packet is missing, it will send a packet resend request to the camera. Packet resends adversely affect performance since bandwidth must be used to transmit the packet again.

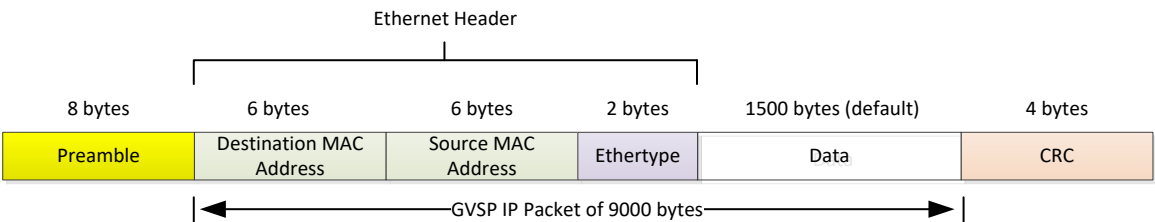
To speed things up, some network cards can "offload" the checksumming, that is, perform the checksum calculation on the network card itself, rather than by the CPU, freeing CPU resources for other tasks.

To reduce CPU load, it is recommended that the offload option be enabled on the NIC for both transmit and receive:

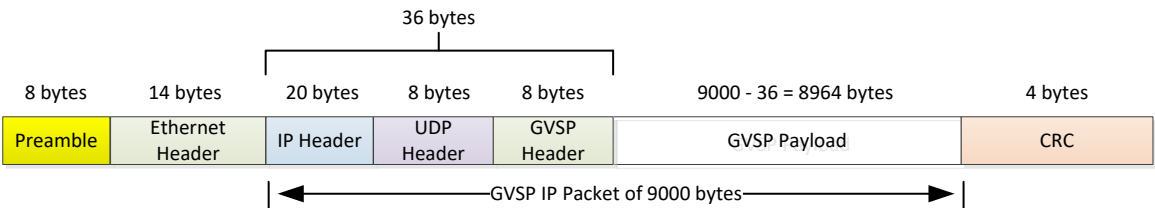


Ethernet Packet Structure

The Ethernet frame or packet provides the addressing (destination and source MAC addresses) and encapsulates the data to send.



For a GVSP IP packet, the structure is as follows:



GigE Vision Control Protocol and GigE Vision Stream Protocol are based on UDP. The Internet Protocol (IP) is the method by which data is sent from one computer to another on a network or across the Internet. Each device must have an IP address to identify that device on the network or on the Internet.

How long to send one packet?

The maximum throughput for Gigabit ethernet is 1000 Mbps (bits/s) or 125 MBps (bytes/s). Therefore, to transfer 1 byte requires 1 clock cycle (1 tick = 0.008 μ s = 8 ns).

However, this throughput does not represent the amount of data sent, since the data (payload) is wrapped in packets (ethernet frame) to send which creates multiple layers of overhead. An interpacket gap is also required (minimum 12 bytes).

For example, for a 9000 byte packet size, when the preamble (8 bytes min), Ethernet header (14 bytes) and CRC (4 bytes) are included the actual packet size is 9026 bytes. The actual amount of image data transferred when using 8-bit pixels is 8964 pixels (9000 GVSP packet size minus 36 bytes of headers (IP, UDP, and GSVP).

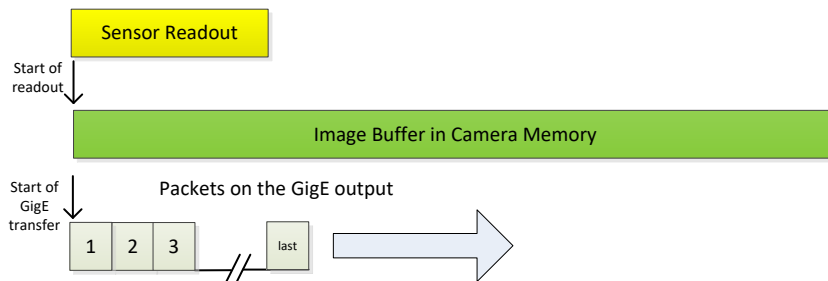
Given the transmission rate of 8 ns, 9026 bytes can be sent in 72.208 μ s.

Sensor Readout and Transmission Rates

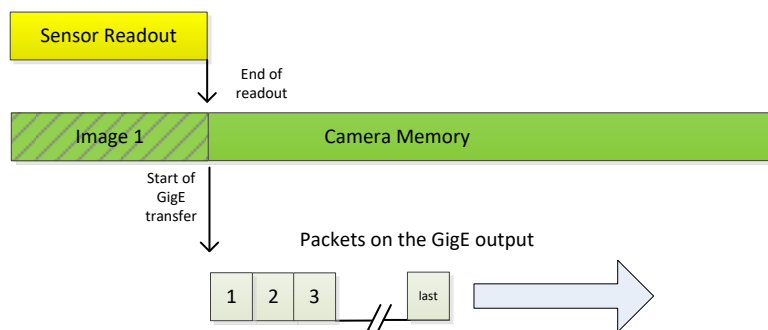
Cameras either assemble and transmit packets during sensor readout, as soon as the number of pixels required by the packet size are available, or buffer the image completely from the sensor and then send the image packets. However, cameras that send packets during sensor readout may also use buffered images if necessary due to a backlog generated by bandwidth limitations or congestion. Depending if the sensor readout is slower than the GigE transmission rate, this can cause apparent fluctuations in transfer rate.

Teledyne DALSA cameras use the first-in/first out model and transmit packets during sensor readout.

Packets Transmitted During Sensor Readout



Packets Transmitted From Buffer After Sensor Readout

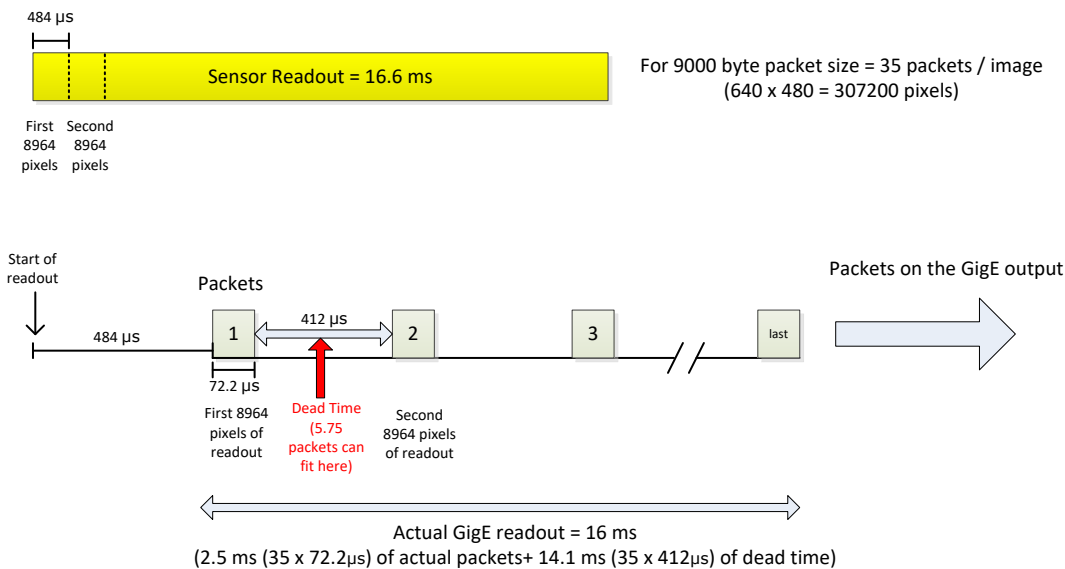


Refer to the camera documentation for details on the sensor readout timing and the packet transmission method. Depending on the camera model, the sensor readout time can be faster than the GigE transmission rate.

For example, the original Teledyne DALSA Genie M640 sensor readout time did not approach the limits of the GigE transmission rate, allowing several cameras to be connected to a single NIC through a switch and not exceed the available bandwidth.

A 640x480 image contains 307200 pixels. When using a 9000 byte packet size, this requires 35 packets per image ($307200 / 8964 = 34.27$). The first packet can be formed and transmitted as soon as 8964 pixels are readout from the sensor ($484\mu\text{s}$ after start of readout). When the sensor readout is complete, the last packet is padded to meet the packet size requirement.

Genie M640: 640 x 480 @ 60 fps



Note: The **Interpacket Delay** feature adds delay to the existing interpacket gap. The GigE Vision standard requires a minimum gap between packets of 96 ns; this cannot be reduced using this feature. This interpacket gap may be larger due to the camera readout limitations. In the example above, the dead time between packets ($412\mu\text{s}$) cannot be reduced, even if the InterPacket Delay is set to 0. The InterPacket Delay increases this interpacket gap, regardless of its size; for example, setting the Interpacket Delay to $10\mu\text{s}$ will increase the actual interpacket gap to $422\mu\text{s}$. If the host system NIC is close to its maximum bandwidth, the interpacket delay can be used to insert time between camera data packets, to prevent packet resend conditions.

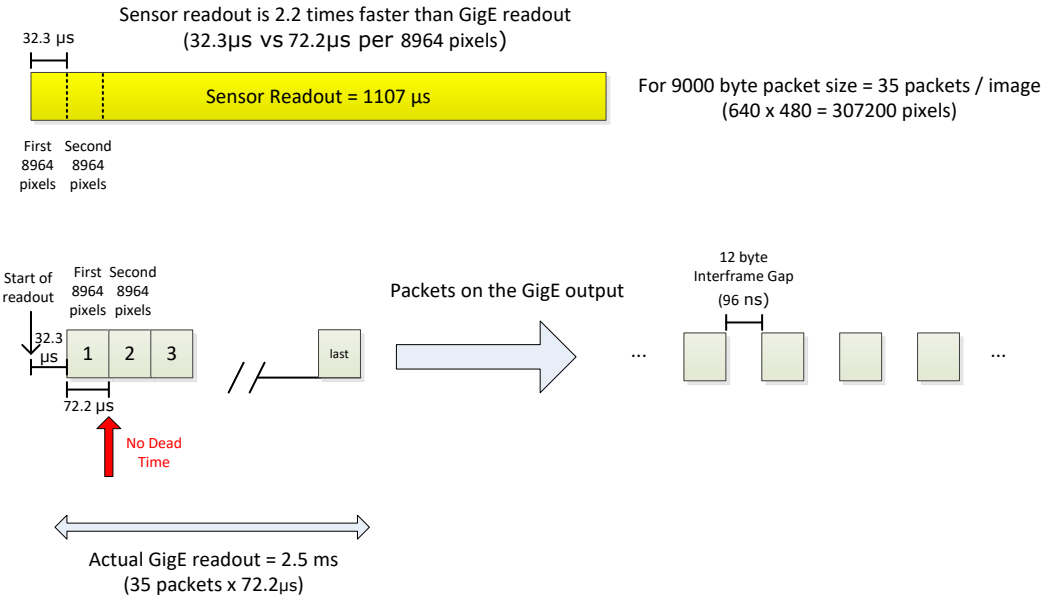
In contrast, the Genie Nano M640 camera sensor readout time is fast enough to exceed the limits of the GigE transmission rate. The GigE Vision standard requires a minimum 96 ns delay between packets to ensure reliable transmission. Therefore, the actual sustainable free-running acquisition frame rate (without dropping frames) over the GigE ethernet connection is lower than the maximum readout rate achievable by the camera sensor.



Note: Certain camera models may use the Device Link Throughput Limit feature to prevent the camera from exceeding the GigE ethernet transmission limit.

The bandwidth calculation must also include the interframe gap (minimum 12 bytes) that separates consecutive packets.

Genie Nano M640: 640 x 480 @ 862 fps



Note: TurboDrive reduces the number of packets required for an image, effectively increasing the maximum frame rate output. With TurboDrive, the packet payload is the same 8964 bytes, but the number of pixels represented by these bytes is greater due to the encoding algorithm. The amount TurboDrive lossless compression achieved is dependent on the type of image.

Managing Network Connections

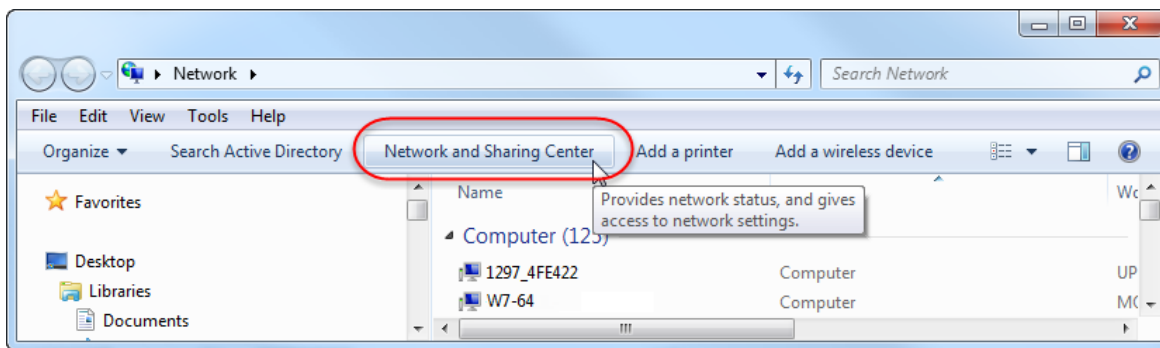
Local area connections make it possible for computers to access resources on the network and the internet. One local area connection is created automatically for each network adapter installed on a computer. This section examines techniques you can use to manage these connections.

Viewing network configuration information

You can view the current configuration for network adapters in several ways. To view configuration settings using the Local Area Connection Status dialog box, follow these steps:

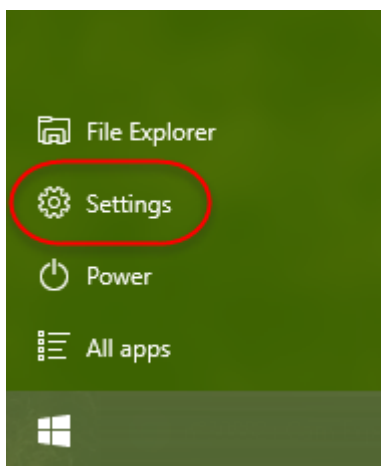
For Windows 7:

- Click Start and then click Network. In Network Explorer, click Network and Sharing Center on the toolbar.

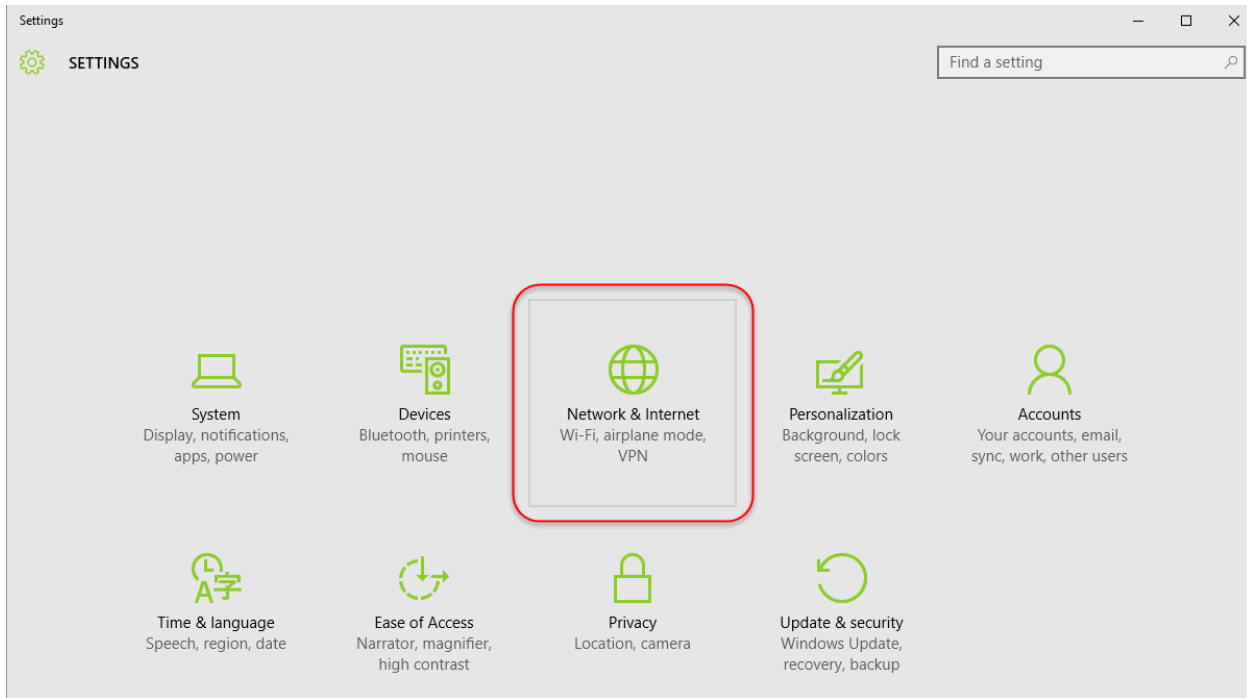


For Windows 10:

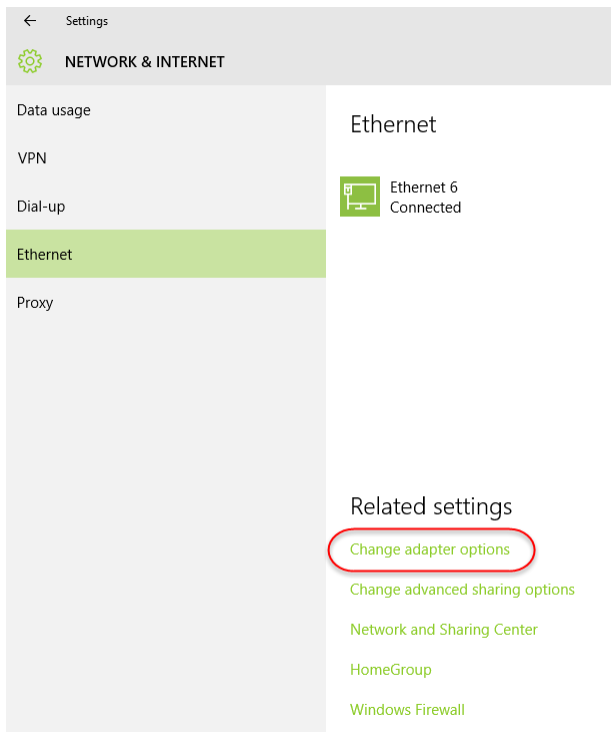
- From the Desktop start menu, click **Settings**.



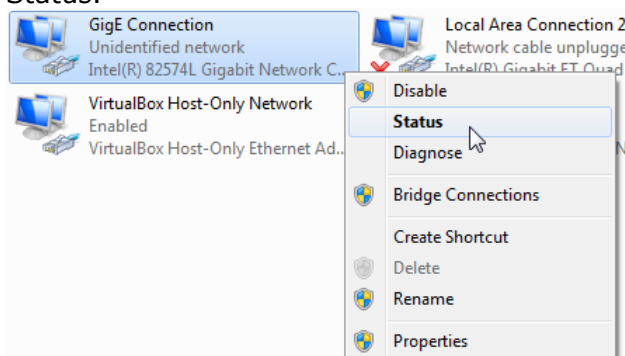
- In the **Settings** dialog, click **Network & Internet**



- In the **Network & Internet** dialog, select **Ethernet** click **Change adapter options**.



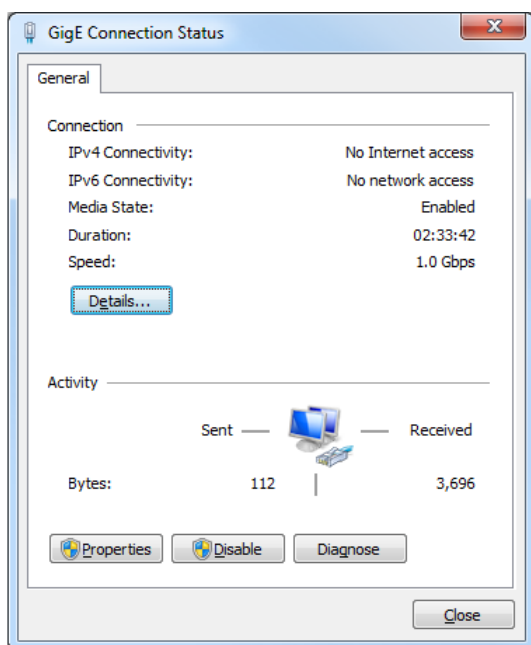
- In Network Connections, right-click the connection you want to work with and then click Status.



This displays the Local Area Connection Status dialog box. If the connection is disabled or the media is unplugged, you won't be able to access this dialog box. Enable the connection or connect the network cable to resolve the problem and then try to display the Status dialog box again.

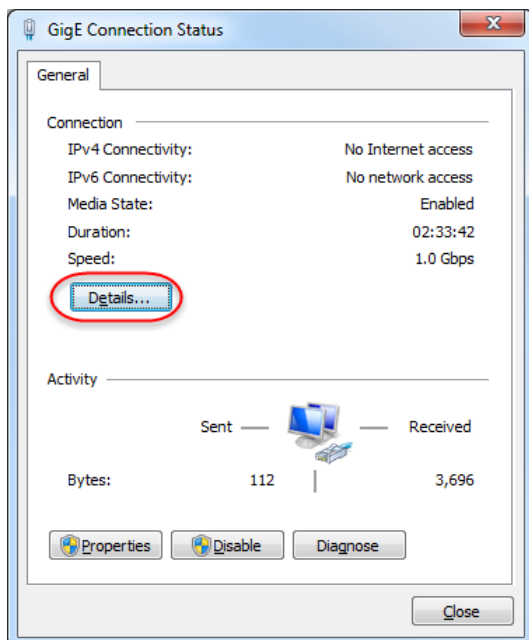
You can rename the connections at any time by right-clicking the connection, selecting Rename, and then typing a new connection name. If a computer has multiple local area connections, proper naming can help you and others better understand the uses of a particular connection.

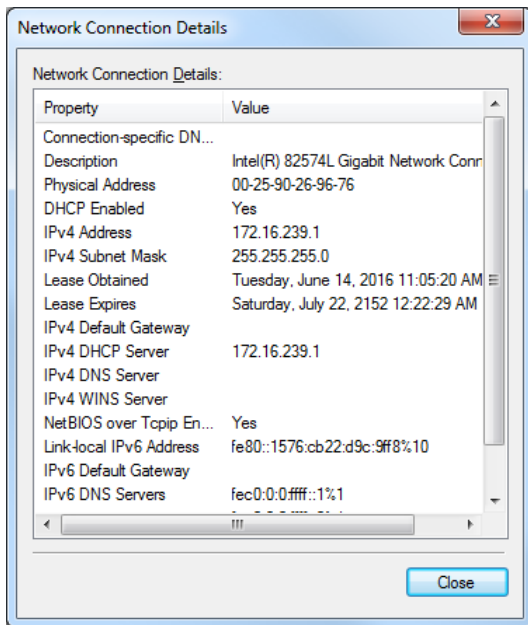
- The General tab of network connection Status dialog box provides useful information regarding the following:



- **IPv4 connectivity** the current IPv4 connection state and type. You'll typically see the status as local when connected to an internal network or not connected when not connected to a network.
- **IPv6 connectivity** the current IPv6 connection state and type. You'll typically see the status as local when connected to an internal network or not connected when not connected to a network.

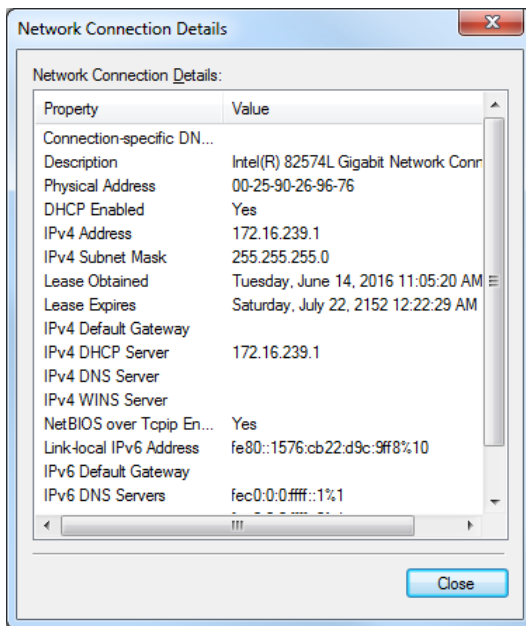
- **Media state** the state of the media. Because the Status dialog box is available only when the connection is enabled, you'll typically see this as enabled.
- **Duration** the amount of time the connection has been established. If the duration is fairly short, the user either recently connected to the network or the connection was recently reset.
- **Speed** the speed of the connection. This should read 10.0 megabits per second (mbps) for 10-mbps connections, 100.0 mbps for 100-mbps connections, and 1 gigabit per second (gbps) for 1-gigabit connections. An incorrect setting can affect the computer's performance.
- **Bytes** the number of bytes sent and the number received by the connection. As the computer sends or receives packets, you'll see the computer icons light up to indicate the flow of traffic.
- Click **Details** to view network connection details.





Details include detailed information about the IP address configuration, such as:

- **Physical address** the machine or media access control (mac) address of the network adapter. This address is unique for each network adapter.
- **IPv4 IP address** the IPv4 address assigned for IPv4 networking.
- **IPv4 subnet mask** the subnet mask used for IPv4 networking.
- **IPv4 default gateways** the IPv4 address of the default gateways used for IPv4 networking.
- **IPv4 DNS servers** IP addresses for DNS servers used with IPv4 networking.
- **IPv4 WINS servers** IP addresses for WINS servers used with IPv4 networking.
- **IPv4 DHCP server** the IP address of the DHCPv4 server from which the current lease was obtained (DHCPv4 only).
- **Lease obtained** a date and time stamp for when the DHCPv4 lease was obtained (DHCPv4 only).
- **Lease expires** a date and time stamp for when the DHCPv4 lease expires (DHCPv4 only).



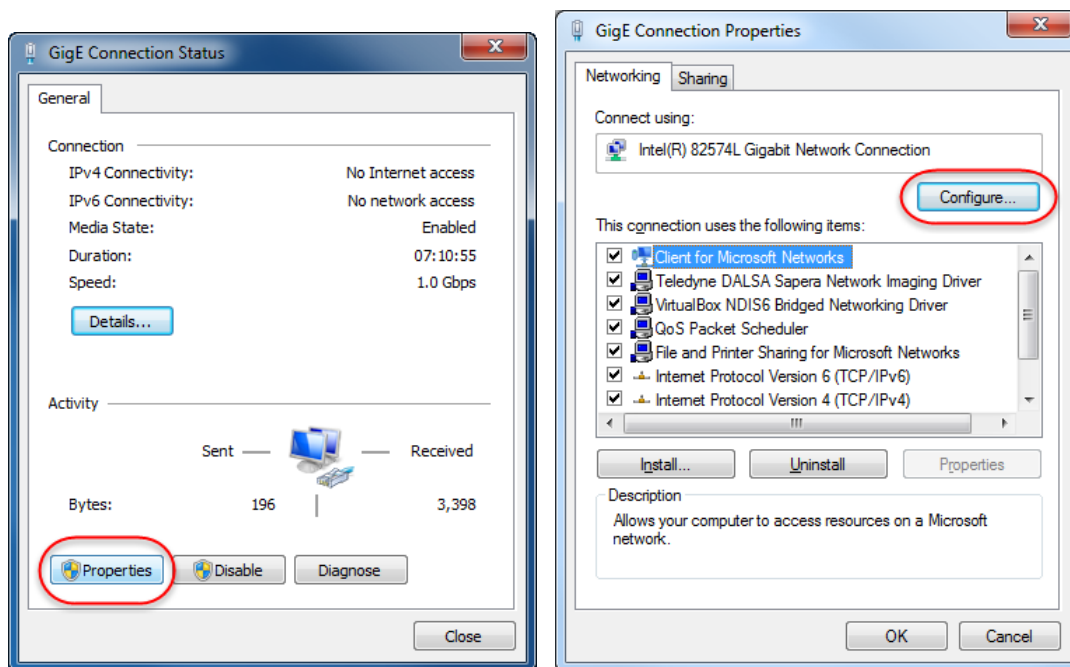
You can also use the IPCONFIG command to view advanced configuration settings. To do so, follow these steps:

1. Click Start and type **cmd** in the Search field.
2. Press Enter.
3. At the command line, type **ipconfig /all** to see detailed configuration information for all network adapters configured on the computer.

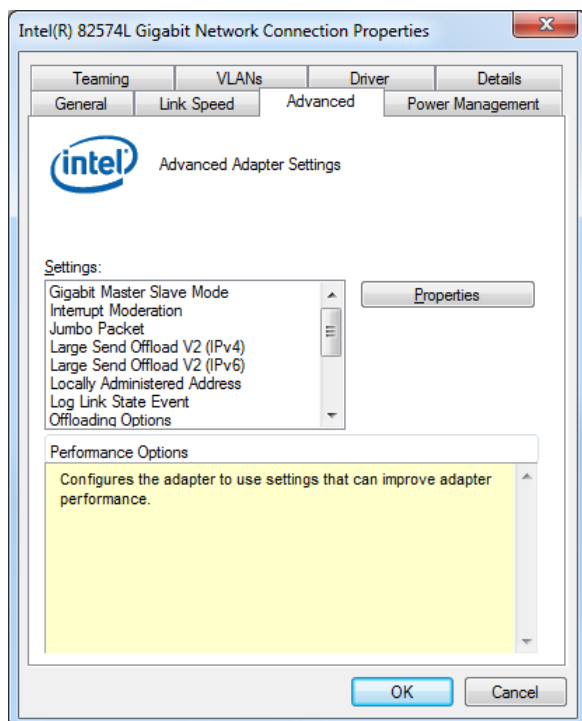
Note The command prompt is started in standard user mode. This is not an elevated command prompt.

NIC Configuration Settings

To access NIC configuration settings, select **Properties** and click the **Configure** button.



The Advanced tab lists the available properties for the NIC.



Network Adapter Optimization

Most Gigabit network interface controllers (NIC) allow user modifications to parameters such as Adapter Buffers and Jumbo Frames. These should be optimized for use with the GigE Vision camera during installation.

Key parameters for network optimization include:

Parameter	Description
Adaptive Inter-Frame Spacing	Enables the network adapter to dynamically adapt to network traffic conditions to avoid excessive Ethernet packet collisions
Flow Control	Flow control mechanism using PAUSE frames.
Interrupt Moderation	Lowers interrupt rate by grouping packets
Jumbo Packets	Jumbo packets minimize CPU usage.
Large Send Offload	Offloads the task of segmenting TCP messages into valid Ethernet frames to the adapter (should be disabled).
Receive Buffers	Sets the number of receive buffers in host memory to store packets.
Receive Side Scaling Queues	Enables multiple CPU cores to manage received Ethernet packets.
Wait for Link	Defines how the driver waits for Auto Negotiation to be successful before reporting the link state

Screenshots for network adapter properties uses the Intel Network Adapter Driver version 21.0. Your specific network adapter may use different dialogs and properties. Depending on the network adapter, different parameters may be available.

	Note that GigE Vision applications will function with 100 mbps networks but may have limited performance, therefore slower NIC adapters are not recommended. If the computer used with the Teledyne DALSA Network Imaging Module does not have an available Gigabit network adapter, a Gigabit NIC needs to be installed in an adapter slot. Typically under Windows the PCI Gigabit NIC is recognized automatically when Windows boots.
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Review the NIC documentation concerning any special driver required for Windows. Install the Gigabit NIC as described by the manufacture documentation.

Network Optimization Types

When optimizing network adapter parameters, the user typically considers one of the following four conditions for the imaging system.



Note: All figures and controls described in this document are from an Intel Gigabit Network Adapter. Other manufacture's network adapters may have similar options and controls.

Quick Response and Low Latency

- Disable Jumbo Packets
- Disable Interrupt Moderation Rate
- Disable Offload TCP Segmentation
- Increase Receive Descriptors
- Disable Receive Side Scaling Queue

Throughput

- Enable Jumbo Packets (value between 1500 and 9000 Bytes might increase the max throughput with certain systems.)
- Disable Flow Control in Ethernet switches (Flow Control can affect throughput with some switches)
- Increase Receive Descriptors
- Enable/Increase Receive Side Scaling Queue
- Interrupt Moderation Rate= Extreme or OFF

System Robustness

- Interrupt Moderation Rate to at least match the number of expected packet/sec
- Enable Jumbo Packets (and use Max values)
- Increase Receive Descriptors
- Enable Flow Control

Lower CPU Utilization

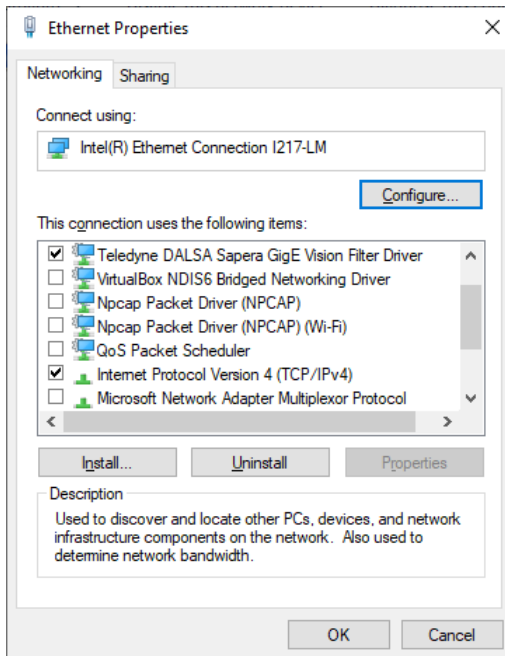
- Low Interrupt Moderation Rate
- Enable Jumbo Packets (and use Max values)
- Increase Receive Descriptors
- Enable Flow Control

Note: This will greatly affect the Maximum GigE throughput

NIC Ethernet Connection Properties

For NICs with GigE Vision cameras connected only the two following options should be enabled:

- Teledyne DALSA Sopera GigE Vision Filter Driver
- Internet Protocol Version 4 (TCP/IPv4)



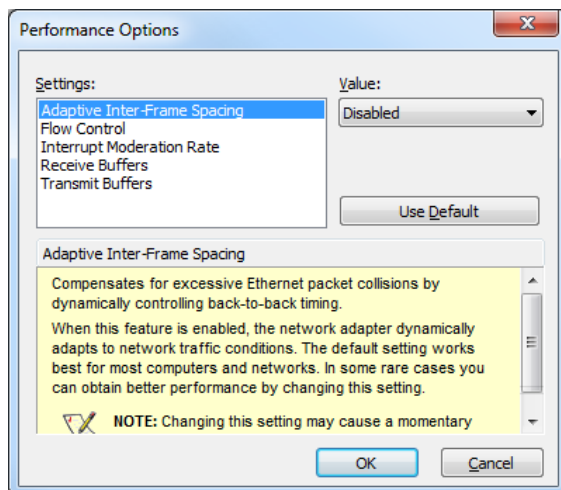
For improved performance, disable the Teledyne DALSA Sopera GigE Vision Filter Driver on any other NIC in the system that is not connected to a GigE Vision device.

Adaptive Inter-Frame Spacing

Adaptive Inter-Frame Spacing (IFS) enables the network adapter to dynamically adapt to network traffic conditions to avoid excessive Ethernet packet collisions. IFS defines the space between Ethernet packets which is a required gap between packets. When this feature is enabled, the network adapter dynamically adapts to network traffic conditions.

If a system NIC is the data sender in a slower half duplex network, allowing a shorter gap would increase the number of Ethernet packets per a specific time period. However, there would be a possibility that the packets are too close together and would not be detected as individual packets, forcing the data to be discarded and the packet resent.

Adaptive Inter-Frame Spacing is disabled by default and should remain so in an imaging system. Cameras streaming images using gigabit networks, especially with buffered Ethernet equipment, will not have any performance improvements.



Flow Control

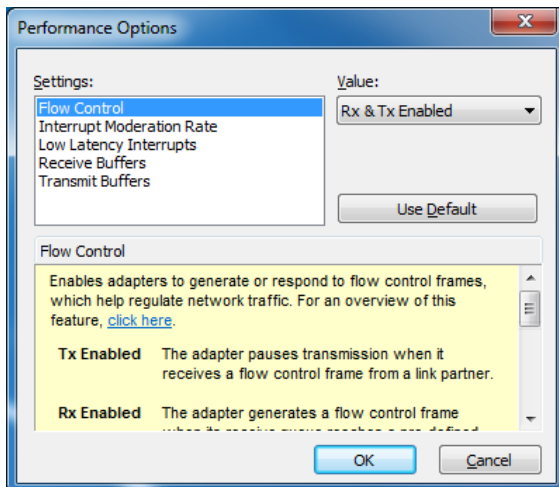
The NIC or Ethernet switch can become overloaded if incoming packets (from multiple cameras) arrive faster than the device can process them. The flow control mechanism eliminates the risk of lost packets and unnecessary packet resend requests that would in turn increase the demand for streaming bandwidth.



Note: It is highly recommended that the Flow Control mechanism be enabled on any NICs or switches to ensure system stability; when disabled, and the system is operating at maximum bandwidth or high CPU usage, in addition to streaming packets, packets on the control channel may be lost such that cameras may not respond to commands.

Enable Flow control on the receive (Rx) queue in the NIC when the imaging network is comprised of multiple cameras connected to a switch which supports PAUSE Frame to manage network traffic. PAUSE frames allows an adapter to send a signal requesting a short pause in frame transmission. The PAUSE frame includes the length of time for the transmitter (in this case, the camera) to stop sending packets. The pause time is measured in units of pause “quanta”, where each unit is equal to 512 bits (4096 ns). The range of possible pause time requests is from 0-65535 (0 effectively cancels any current PAUSE frame request).

The downside to managed network traffic is that the PAUSE Frame control will reduce the absolute maximum transfer bandwidth possible on the network.



Typically, possible values are:

- **Tx Enabled:** The adapter pauses transmission when it receives a flow control frame from a link partner.
- **Rx Enabled:** The adapter generates a flow control frame when its receive queue reaches a pre-defined limit.
- **Rx & Tx Enabled:** The adapter generates and responds to flow control frames.
- **Disabled:** The adapter does not generate or respond to flow control frames.



Refer to the switch documentation for information on enabling the flow control mechanism (for example, for certain switches the Link Aggregation option can disactivate flow control).

Interrupt Moderation Rate

For Gigabit ethernet, the maximum packet rate is extremely high, and handling one interrupt per packet can be very inefficient. Interrupt moderation allows the NIC to raise one interrupt for a group of packets.

When a packet arrives the adapter generates an interrupt which allows the driver to handle the packet. At greater link speeds, more interrupts are created, and CPU utilization also increases. This results in poor system performance. When you enable Interrupt Moderation, the interrupt rate is lower, and the result is better system performance.

Interrupt moderation rate sets the rate at which the controller moderates or delays the generation of interrupts making it possible to optimize network throughput and CPU utilization. The default setting (Adaptive) adjusts the interrupt rates dynamically depending on traffic type and network usage.

Choosing a different setting may improve network and system performance in certain configurations.

Without interrupt moderation, CPU utilization increases at higher data rates because the system must handle a larger number of interrupts. Interrupt moderation causes the network driver to accumulate interrupts and send a single interrupt rather than a series of interrupts. At higher data rates, a high interrupt moderation setting may improve system performance. At low data rates, a lower interrupt moderation setting is preferred as delayed interrupts causes latency.

To reduce the number of interrupts, many NICs use interrupt moderation. With interrupt moderation, the NIC hardware will not generate an interrupt immediately after it receives a packet. Instead, the hardware waits for more packets to arrive, or for a time-out to expire, before generating an interrupt. The hardware vendor specifies the maximum number of packets, time-out interval, or other interrupt moderation algorithm.

The measured round-trip time for a packet is one of the most commonly used techniques to determine the network bandwidth between two endpoints. However, when interrupt moderation is enabled, receiving a packet does not generate an immediate interrupt and therefore the perceived round-trip time for a particular packet becomes larger than the average time. To allow accurate measurement of round trip time for a packet, NDIS provides the ability to disable and enable interrupt moderation on demand.

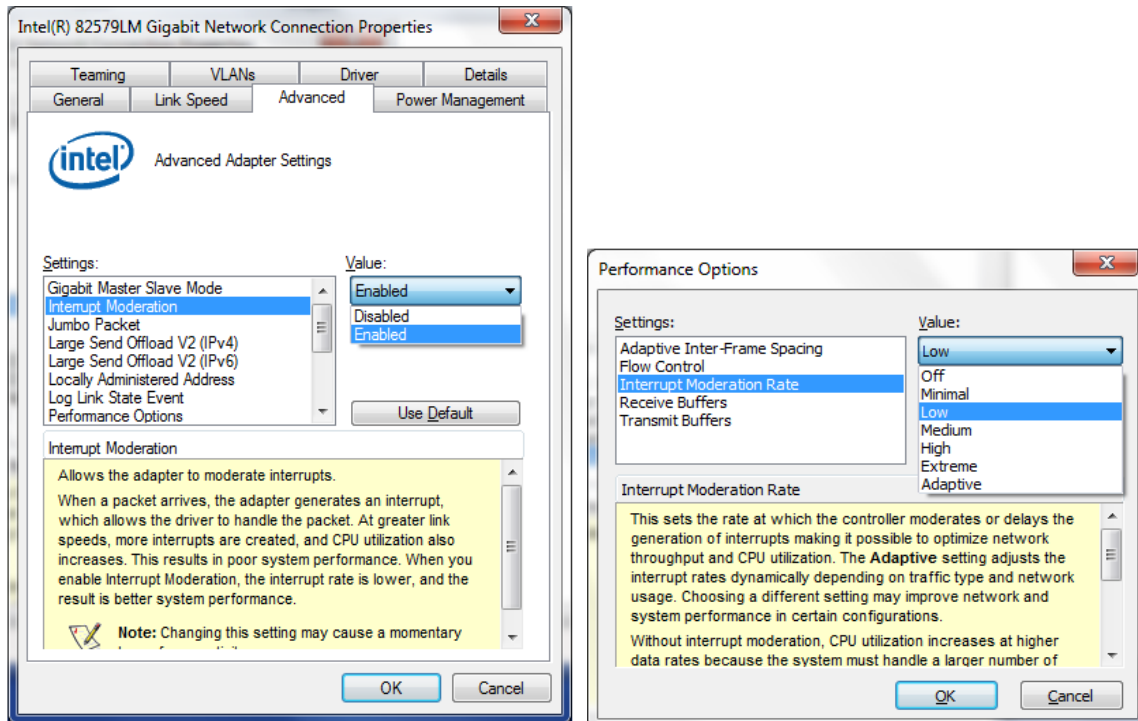
The default setting is optimized for common configurations. Changing this setting can improve network performance on certain network and system configurations.

When an event occurs, the adapter generates an interrupt, which allows the driver to handle the packet. At greater link speeds, more interrupts are created, and CPU rates also increase. This results in poor system performance. When you use a higher ITR setting, the interrupt rate is lower, and the result is better system performance.

Interrupt Moderation Rate Optimization

Typically, by default the NIC driver set the interrupt moderation rate to 'Adaptive' where the interrupt rate automatically balances packet transmission interrupts and host CPU performance. In most cases no manual optimization of the Interrupt Moderation Rate parameter is required.

In some conditions, video frames from the GigE Vision camera may be transferred to host display or memory buffer as data bursts instead of a smooth continuous stream. The NIC may be over moderating acquisition interrupts to avoid over-loading the host CPU with interrupts. If priority is required for acquisition transfers (i.e. a more real-time system response to the camera transfer), the moderation rate should be reduced by manually adjusting the NIC parameter.



The Interrupt Throttle Rate (ITR) is the rate at which the controller moderates interrupts. For example, when set to High, up to 2000 interrupts per second are suppressed by the NIC.

- Change the value from the default 'Adaptive'. Try different values from 'Off' to improve the real-time GigE Vision camera acquisition response relative to the over-all host computer usability. Note that no interrupt moderation may make the host computer seem unresponsive to other applications.
- Change the value to 'Extreme' will minimize CPU utilization. However this will add a small latency when applications are notified that an image is ready to process.

Possible values are:

- **Adaptive** — (ITR = -1): No interrupts/sec defined. Dynamically changed. Not recommended but it is the default for most systems.
- **Off** — (ITR = 0): No limit. Recommended to maximize real-time performance.
- **Minimal** — (ITR = 200 interrupts/sec)
- **Low** — (ITR = 400 interrupts/sec)
- **Medium** — (ITR = 950 interrupts/sec)
- **High** — (ITR = 2000 interrupts/sec)
- **Extreme** — (ITR = 3600 interrupts/sec): Can be used to minimize CPU usage.



A higher ITR rate also means the driver has more latency in handling packets. If the adapter is handling many small packets, lower the ITR so the driver is more responsive to incoming and outgoing packets.

Jumbo Packets

Jumbo packets, also known as jumbo frames, are larger than standard Ethernet frames, which are 1500 bytes in size. If large packets make up the majority of traffic and more latency can be tolerated, Jumbo Frames can reduce CPU utilization and improve wire efficiency. With good gigabit Ethernet connections with minimal packet resend conditions, host computer performance can be further improved by increasing the data packet size.

Each streaming video packet(s) causes an interrupt in the host computer. Therefore increasing the packet size reduces the CPU usage percentage required to handle video data from the GigE Vision camera. Jumbo packets provide greater efficiency in data transmission, since each frame carries more user data with the same fixed protocol overhead and underlying packet delays. Use of jumbo packets results in fewer Ethernet packets being sent across the network. Fewer packets results in less host system CPU processing providing greater system throughput. Overall, a quality network should support a higher continuous transfer rate with a packet size larger than the default (1500 bytes).

While large Ethernet packets provide greater efficiencies, they also cause greater lag time and latencies over the network. Any packet resends have a larger impact in average throughput. In networks with Ethernet switches using Flow Control, the largest jumbo packet may not produce the best transfer bandwidth. Large packets may also fill buffer queues within network equipment between the camera and host computer. Multiple cameras connected to an Ethernet switch, typically require the switch to support the PAUSE frame control, which in turn implies that the network equipment has sufficient buffers to maintain continuous camera acquisitions.



Note: The actual size of the packet (in bytes) to get an optimal throughput is system depends (NIC configuration, Host system load and the Ethernet switch internal configuration). A value between 1500 and 9000 Bytes might increase the max throughput with certain system.

Usage Considerations

Enable jumbo packets only if devices across the network support them and are configured to use the same frame size. When setting up jumbo packets on other network devices, be aware that different network devices calculate jumbo packets sizes differently. Some devices include the header information in the frame size while others do not. Intel adapters do not include header information in the frame size.

When configuring jumbo packets on a switch, set the frame size 4 bytes higher for CRC, plus 4 bytes if using VLANs or QoS packet tagging.

A Virtual Local Area Network (VLAN) is a flexible arrangement where computers connected via a VLAN Ethernet switch are not necessarily on the same LAN broadcast domain. Refer to the VLAN Ethernet switch documentation for implementation and configuration details.

Verifying the Jumbo Packet Setting

The screenshot below shows that the GigE Vision camera was found and there is no IP conflict. Note that an application must connect to the GigE Vision camera first.

The Maximum Packet Size field is highlighted in yellow, indicating that jumbo packets are not enabled on the NIC used with the camera or that the control application is forcing a safe packet size. Using a larger packet size improves host CPU performance, which can be critical when using multiple cameras.

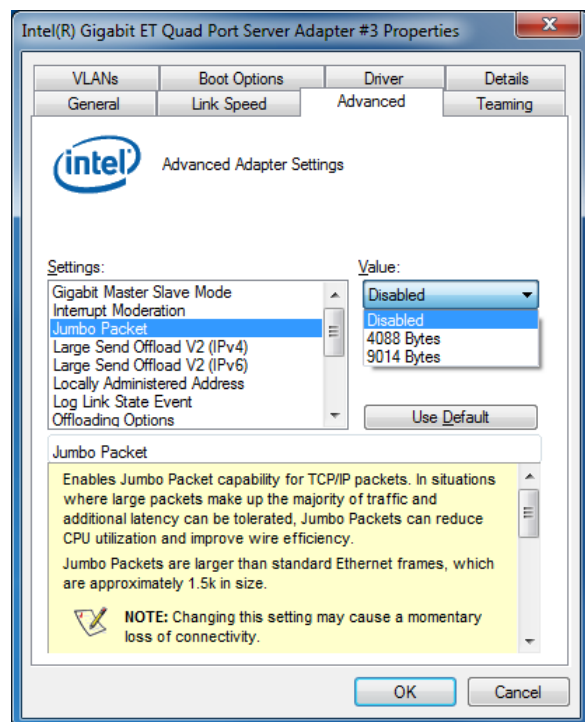
[illegible]

Note that the Maximum Packet Size field is updated only when an application such as CamExpert communicates with the camera.

Jumbo Packet Property

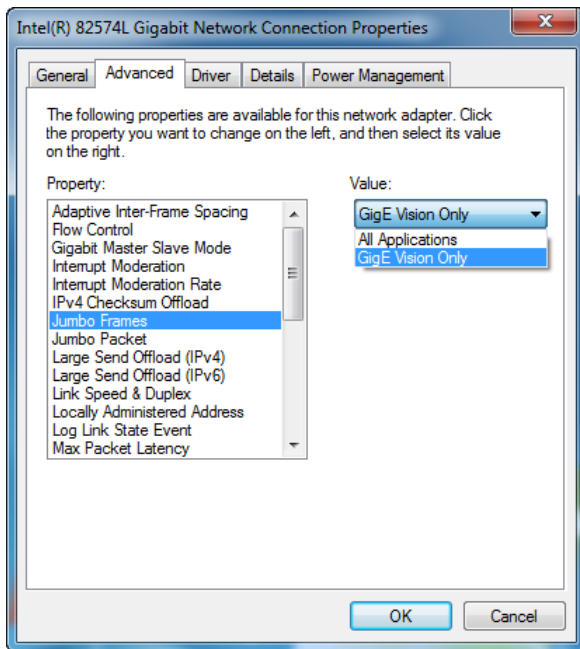
The Jumbo Packet property allows you to configure the jumbo packet size. When set to the maximum size, the GigE Vision camera can then be configured to use its maximum packet size.

The user must evaluate different packet size values to maximize transfer bandwidth. The default or the largest packet size may not produce optimum transfers.



Jumbo Frames Property

Certain NICs also expose the Jumbo Frames property, which allows jumbo packets to be enabled for all applications or for GigE Vision applications only.



Third Party Technical Reference

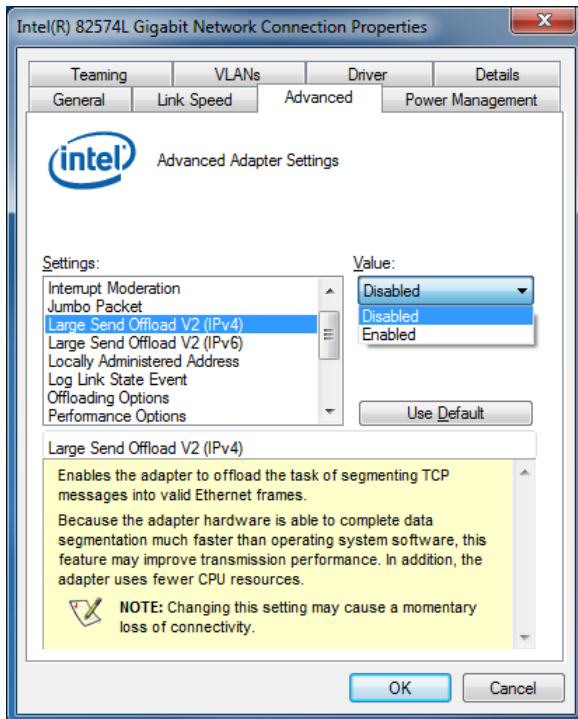
The paper presented by the Ethernet Alliance provides a technical resource when determining the need for jumbo packets.

<http://www.ethernetalliance.org/wp-content/uploads/2011/10/EA-Ethernet-Jumbo-Frames-v0-1.pdf>

Large Send Offload

The Large Send Offload (LSO) is used on the host side to allow the TCP/IP network stack to build a large message of up to 64KB before sending it to the Ethernet adapter, which then segments it into smaller data packets.

In general, for most GigE Vision applications, this feature should be disabled. The LSO mechanism does not support dynamic frame sizes therefore if network components do not all have the same maximum transmission unit (MTU) packets can be dropped and adversely affect network performance.



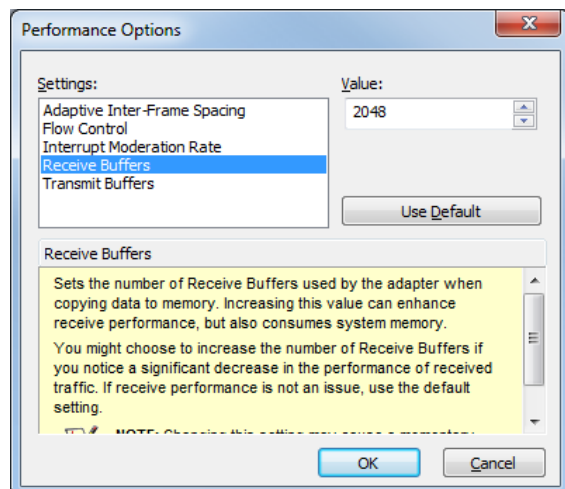
Receive Buffers

Receive buffers are used by the adapter when copying data to memory. Receive buffers (also referred to as adapter buffers) are allocated in system memory by the NIC driver. Receive buffers are the same size as the packet size.

When the host computer CPU is busy with tasks other than the imaging application, incoming image packets remain in the PC memory buffers allocated to store packets instead of immediately being copied into the image application buffers.

By increasing the NIC host buffers, more incoming image packets can be stored by the NIC before it must start discarding them. This provides more time for the PC to switch tasks and move image packets to the image buffer. Increasing this value can enhance receive performance, but also consumes system memory.

You might choose to increase the number of Receive Buffers if you notice a significant decrease in the performance of received traffic. If receive performance is not an issue, use the default setting.



Not all network boards allow increasing their buffer count and even among those that do, such as the Intel NIC, different versions will have different maximum values.

Receive Descriptors Optimization

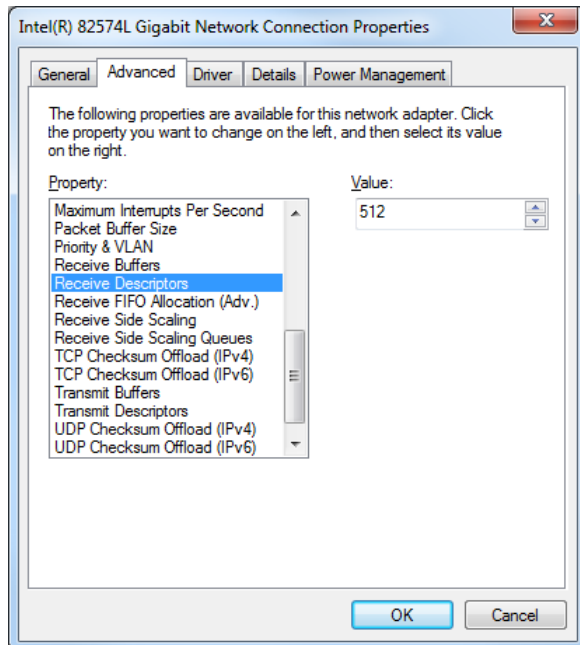
Some NICs may provide access to the receive descriptor property. Not all NICs expose the receive descriptor property, and different NIC may have different maximum receive descriptor values. Refer to the NIC user documentation for details on configuring this parameter.

Receive descriptors are data segments that enable the adapter to allocate received packets to memory. A receive descriptor is a data structure that contains the receive data buffer address and fields for hardware to store packet information. Memory buffers pointed to by descriptors store packet data. Descriptors store information about the buffers such as the physical address, the length, status and command information about the referenced buffers.

Increasing this value can enhance the receive performance, but also consumes system memory; each descriptor uses 2 KB of memory. Depending on the size of the packet, each receive packet requires one or more Receive Descriptors.

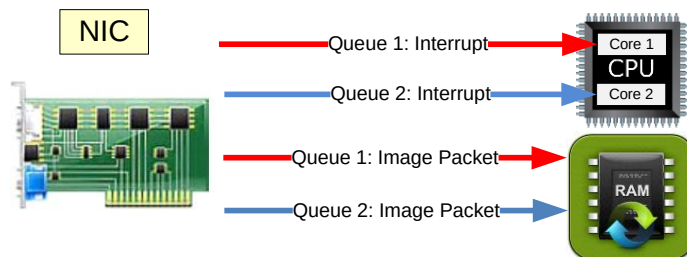
The more descriptors available the more packets can be assigned to memory on the host.

It is recommended to change the value to the largest value supported by the installed NIC.



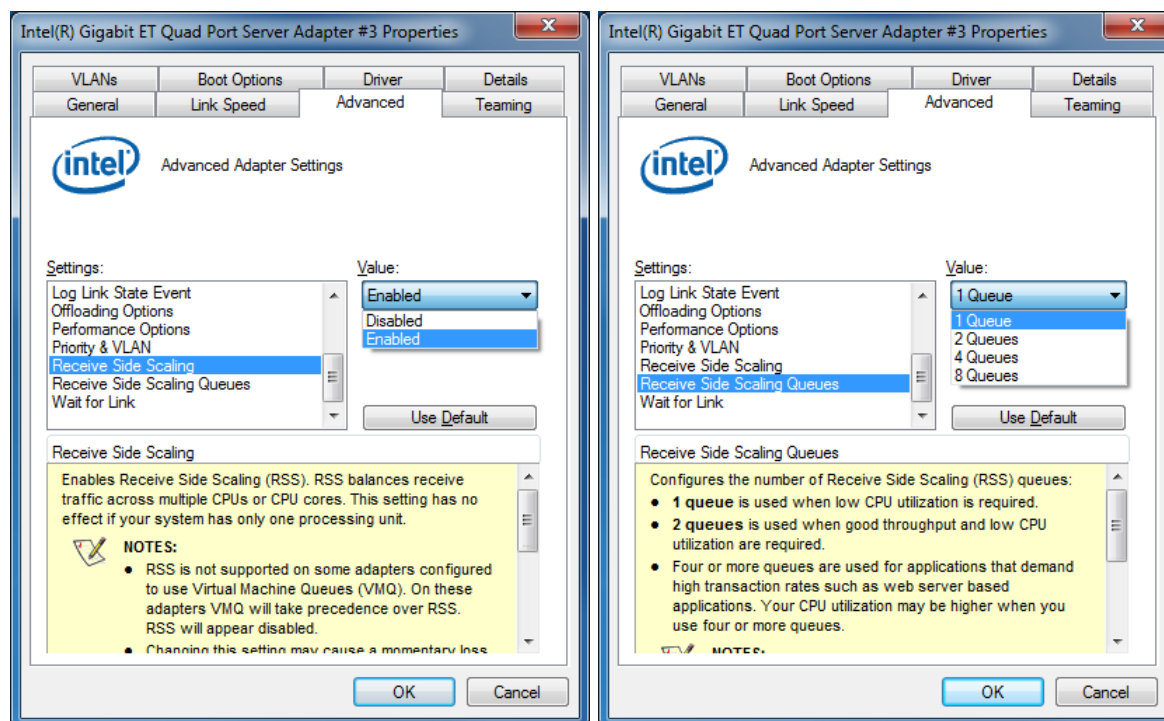
Receive Side Scaling Queue

Receive Side Scaling (RSS) Queue, if supported, enables multiple CPU cores, to manage received Ethernet packets, which improves system cache utilization. Received packets are assigned to one of the receive queues. Each queue uses a separate interrupt and is routed to a different CPU/core.



If your NIC supports Receive Side Scaling Queues, consider enabling this feature as per the following guidelines which affect receive throughput and latency:

- 1 queue is used when low CPU utilization is required.
- 2 queues is used when good throughput and low CPU utilization are required.
- Four or more queues are used for applications that demand high transaction rates such as web server based applications. Your CPU utilization may be higher when you use four or more queues.

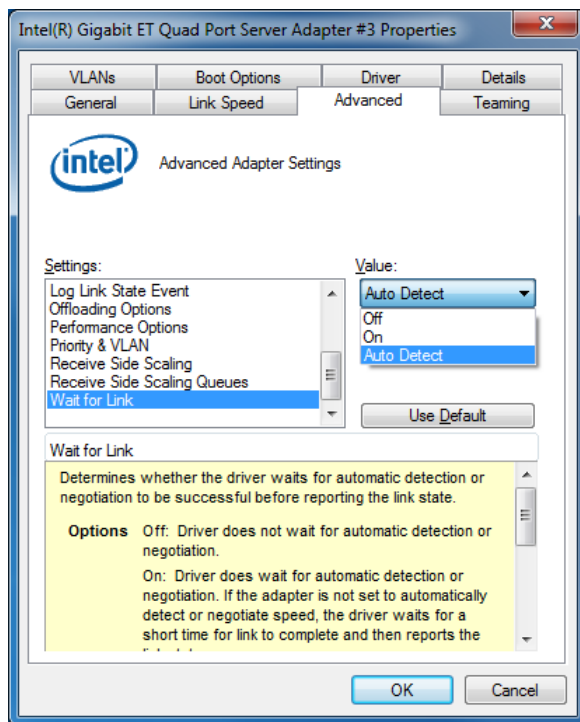


Wait for Link

The Wait for Link property defines how the driver waits for Auto Negotiation to be successful before reporting the link state. Auto Detect is recommended for camera applications.

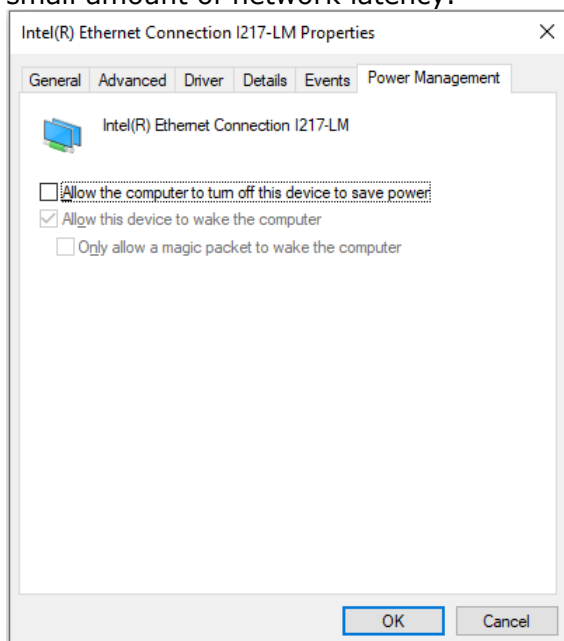
Typical adapter options are:

- **Auto Detect:** Automatically set to On or Off depending on speed and adapter type when the driver is installed.
- **Off:** Driver does not wait for Auto Negotiation
- **On:** Driver waits for Auto Negotiation. If the speed is not set to Auto Negotiation, the driver waits for a short time for link to complete and then reports the link state.

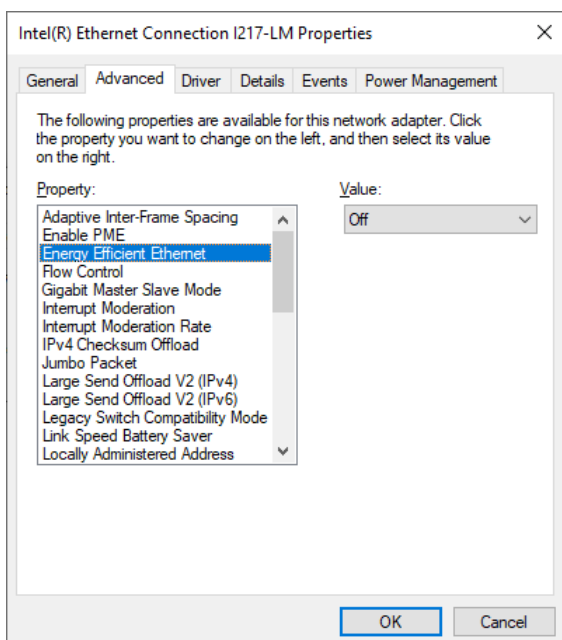


NIC Power Management

Any NIC power management features should be turned off since these transitions may introduce a small amount of network latency.



This includes features such as Energy Efficient Ethernet.



Camera Features Controlling Image Throughput

Certain GigE camera features can limit the bandwidth available for image transmission. This is normally used when multiple cameras share the same GigE link. For example, with Teledyne DALSA cameras such as the Genie Nano, the GigE Vision Transport Layer category contains features such as the Device Link Throughput Limit that can be disabled and the Interpacket Delay that can be reduced. Refer to the camera documentation for more information on relevant features.

Parameter	Value
Device Link Selector	0
Device Link Throughput Limit	On
Device Link Throughput Limit (in Bps)	115000000
Stream Channel Selector	0
Device Link Speed (in Mbps)	1000
PacketSize	9000
Interpacket Delay	7

Device Link Throughput Limit

Limits the maximum data bandwidth streamed by the device.

Packet Resend Buffer Size

Specifies the amount of memory to reserve, in MB, for the packet resend buffer.

Interpacket Delay

Specifies the delay to insert between each packet on the selected stream channel.

Interpacket Delay

When using smaller packets, the interpacket delay (*GevSPCD* feature) has a correspondingly larger effect since more packets introduce more interpacket delays. Depending on the size of the interpacket delay this effect may have an impact on available bandwidth.

To modify the interpacket delay (*GevSPCD* feature), the Device Link Throughput Limit feature must be disabled (*DeviceLinkThroughputLimitMode* = *Off*). The *DeviceLinkThroughputLimit* feature reduces the maximum transmission rate of the GigE link by automatically calculating the required interpacket delay for a given transmission rate. The Device Link Throughput Limit is enabled by default (*DeviceLinkThroughputLimitMode* = *On*).

The maximum throughput for Gigabit ethernet is 1000 Mbps (bits/s) or 125 MBps (bytes/s), however in practice the usable bandwidth for reliable image transfers is lower due to network overhead (packet headers/footers, control packets, and so forth). The default 7 microsecond interpacket delay provided by the Device Link Throughput Limit reduces available bandwidth by 875 bytes per packet (1 byte requires 1 clock cycle (1 tick = 0.008 μ s = 8 ns). If 12000 packets are sent per second, the available bandwidth is lowered by $\sim$ 10 MB/s. This ensures reliable transmission by providing a buffer between the theoretical maximum GigE transmission rate of 125 MBs, reducing the possibility of dropped frames due to network overhead. To avoid reaching the limits of the network capabilities which can result in lost frames, it is recommended that caution be used when adjusting the interpacket delay so as to ensure an adequate buffer exists to accommodate for variations in network overhead.

Note, the more cameras and hubs on the network, the greater the network overhead and the lower the actual bandwidth available.

Host Computer System Considerations

Disable Anti-Virus Software

Anti-virus software can reduce overall system performance of a Windows system. This can affect the handling of streaming packets from NICs.

If anti-virus software is disrupting system performance, it is recommended to uninstall the software or permanently disable it; refer to the anti-virus documentation for information on how to permanently disable the software.



Note: Depending on the software, even when disabled, the antivirus will re-activate automatically after you restart your device; verify the anti-virus behavior to ensure that it is in fact permanently disable to avoid this.

Processor Power Management Options

6th generation Intel core processors (also known as Skylake) include processor power management (PPM) settings that can adversely affect imaging applications. These settings are designed primarily for energy saving purposes and can reduce the CPU performance.

The following parameter settings are recommended:

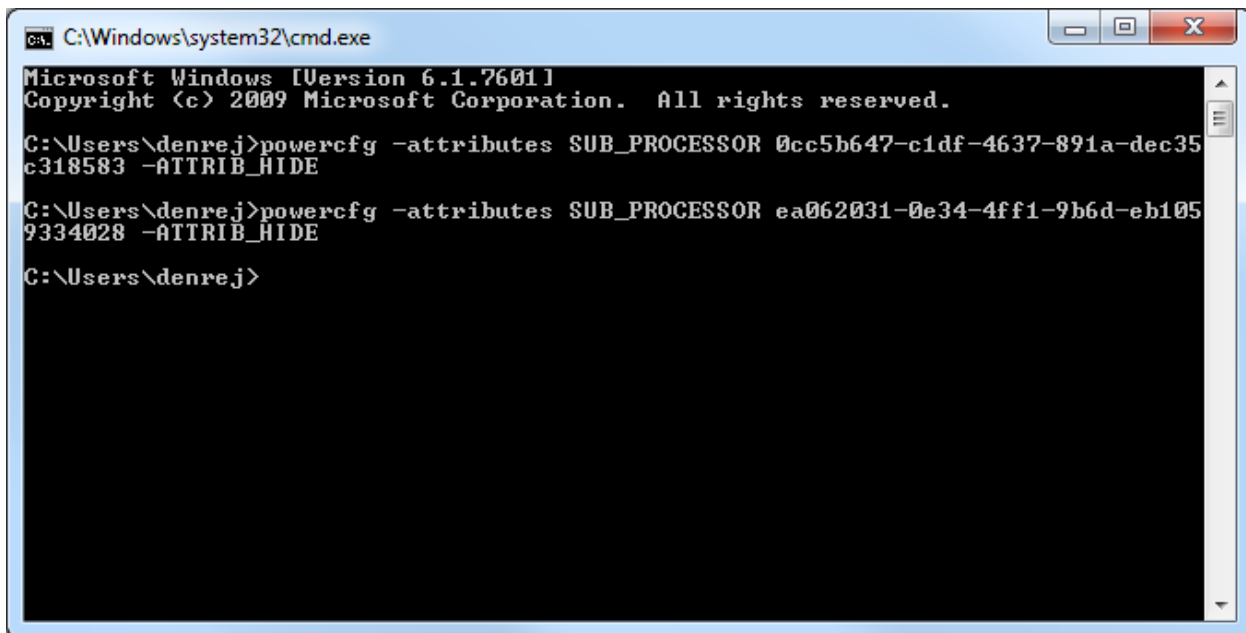
Parameter	Description
Minimum Processor State	Determines the range of CPU performance states (P-states). The number of P-states varies by processor. In effect, this will vary the processor's clock speed to meet workload requirements or decrease it to reduce power consumption.
Processor performance core parking min cores	Core parking reduces the number of logical processors in use.

Enabling Hidden Parameters

By default, certain PPM parameters are hidden. To make them accessible the following commands must be issued using the command prompt:

```
powercfg -attributes SUB_PROCESSOR 0cc5b647-c1df-4637-891a-dec35c318583 -ATTRIB_HIDE
```

```
powercfg -attributes SUB_PROCESSOR ea062031-0e34-4ff1-9b6d-eb1059334028 -ATTRIB_HIDE.
```



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

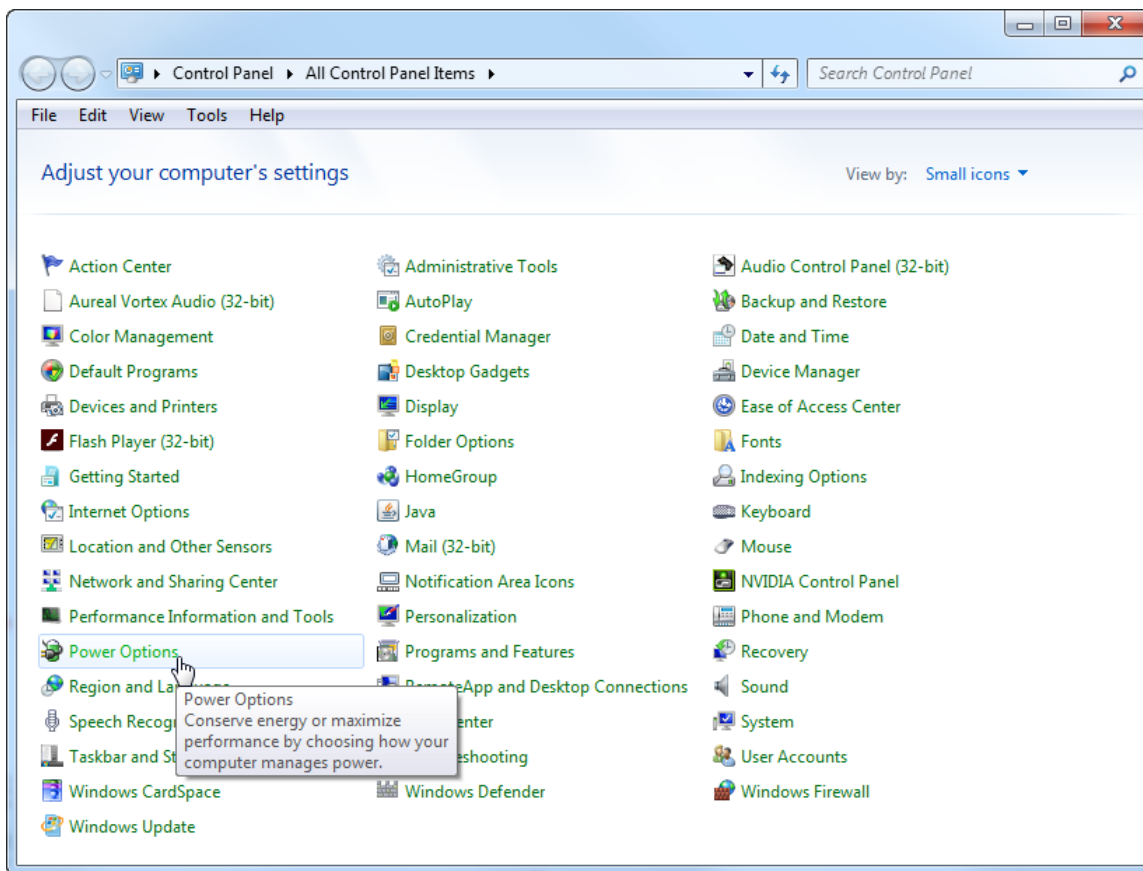
C:\Users\denrej>powercfg -attributes SUB_PROCESSOR 0cc5b647-c1df-4637-891a-dec35
c318583 -ATTRIB_HIDE

C:\Users\denrej>powercfg -attributes SUB_PROCESSOR ea062031-0e34-4ff1-9b6d-eb105
9334028 -ATTRIB_HIDE

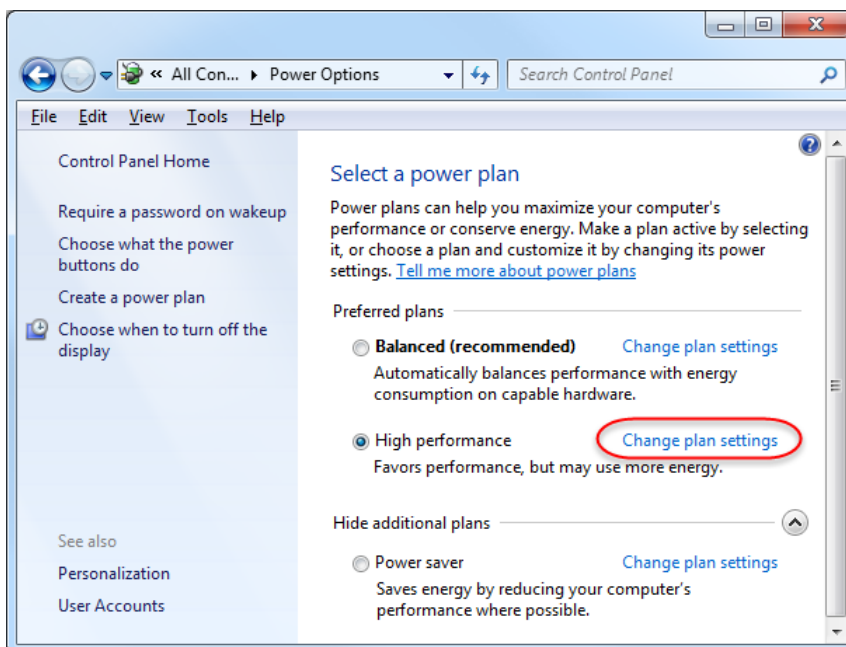
C:\Users\denrej>
```

Changing Processor Power Management Settings

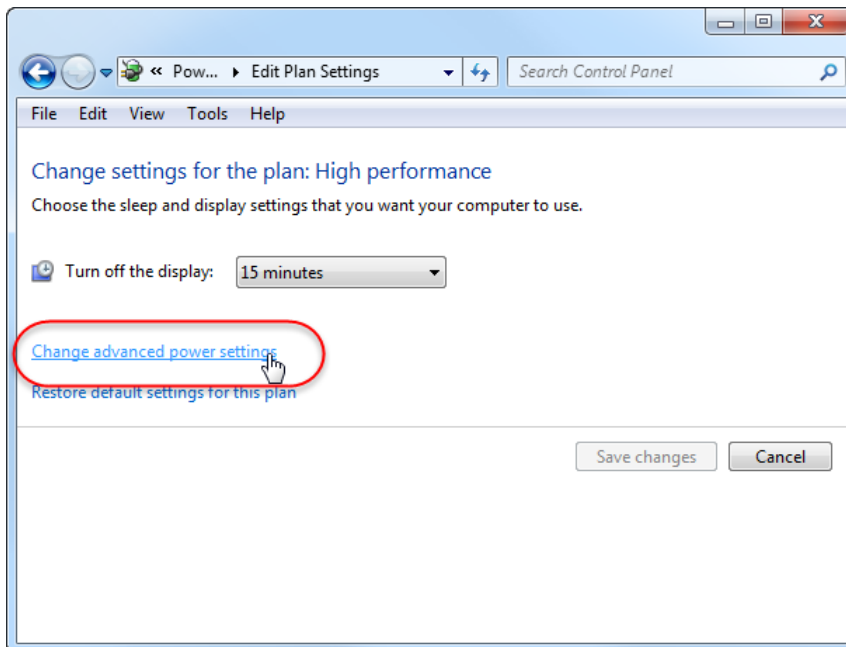
To change PPM settings, from the Control Panel, select **Power Options**.



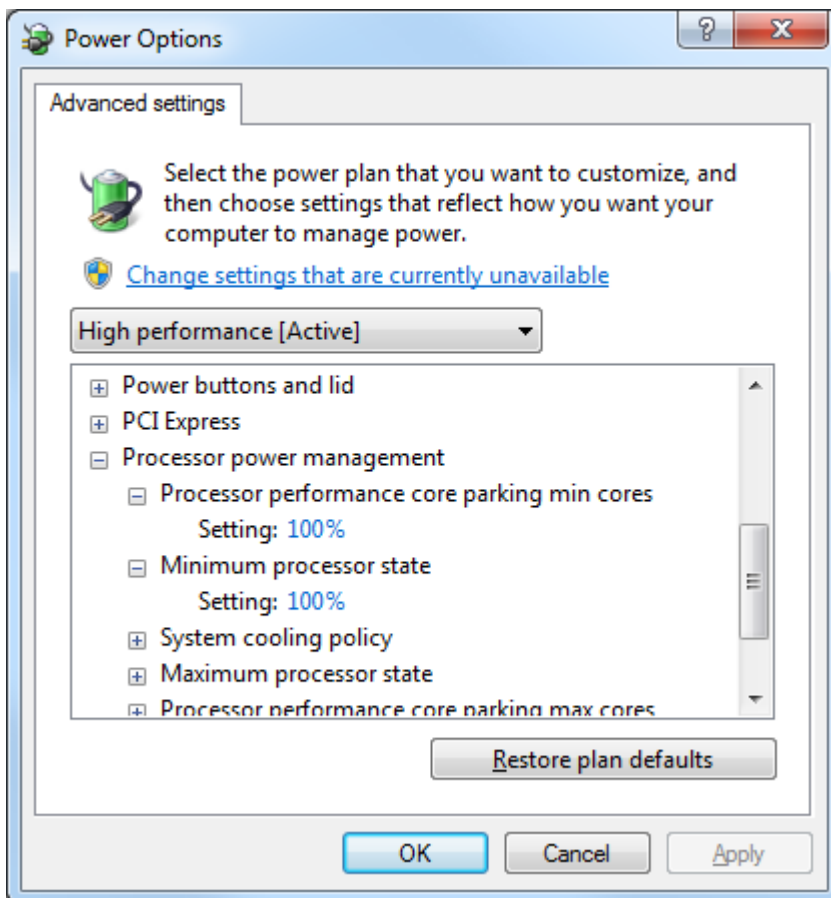
Use the **High Performance** option and click **Change plan settings**.



Click **Change advanced power settings**.



The parameter **Processor performance core parking min cores** should be set to 100% to avoid the shutdown of cores. The **Minimum processor state** should be set to 100%.



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